Physics 12 Winter 2026
Week 9 Discussion

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1. **Summary of Topics from Last Week**

   - *From the lectures*
     - ◇ Electromagnetic waves, Energy of a photon $E = h\nu = hc/\lambda$
     - ◇ Radiation intensity $I = P/A$
     - ◇ Revisited: black body spectrum, Stefan-Boltzmann law $P = A\sigma T^4$
     - ◇ Wien's displacement law $\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m \cdot K}$
     - ◇ Earth's atmosphere, UV from sunlight
     - ◇ Solar constant $S = 1360\ \mathrm{W/m^2}$, Insolation
     - ◇ Transparency/opacity
     - ◇ Greenhouse effect
     - ◇ Solar heating (passive/active)
     - ◇ Thermal power systems, Stirling engine
     - ◇ Solar energy payback time
     - ◇ Photoelectric effect
     - ◇ Photovoltaic (PV) cell
     - ◇ Energy bands (valence band, conduction band)
     - ◇ Semiconductor, doping (p-type, n-type)
     - ◇ PN junction
     - ◇ PV cell/solar panel structure, efficiency
     - ◇ Economics of solar power
     - ◇ Wind power $P/A = \frac{1}{2}\rho v^3$
     - ◇ Energy payback time
     - ◇ Hydroelectric power
   - *From the textbook* (Wolfson 4th edition)
     - ◇ Solar constant, Insolation, Solar spectrum (Chapter 9.1)
     - ◇ PV cells, PN junction (Chapter 9.2)
     - ◇ Passive and active solar heating, solar collectors (Chapter 9.3)
     - ◇ Solar thermal power system, power towers (Chapter 9.4)

- ◇ Environmental impacts, energy payback time (Chapter 9.5-6)
- ◇ Hydropower, hydrologic cycle, transpiration (Chapter 10.1)
- ◇ Wind power (Chapter 10.2)
- ◇ Other indirect solar energy sources (Chapter 10.3-4)

2. **Brain-warmer/mock quiz.**

(a) **Wind power.** The density of air under normal conditions is about 1.2 kg/m^3. For a wind speed of 10 m/s and a wind turbine with a blade area of 10 m^2, what's the maximum extractable wind power?

Hint: the Betz limit is $e_{\text{Betz}} = 16/27 = 0.59$.

A. 2.18 kW

B. 3.54 kW

C. 5.90 kW

D. 6.00 kW

E. 7.08 kW

Answer: B. The total available wind power is

$$\frac{1}{2}\rho v^3 A = 0.5(1.2 \text{ kg/m}^3)(10 \text{ m/s})^3(10 \text{ m}^2) = 6 \text{ kW} \tag{1}$$

Multiply this by the Betz limit to get the maximum extractable power: 0.59×6 kW $= 3.54$ kW.

(b) **Sunburn.** On a clear, sunny day in San Diego, you can easily get sunburn outdoors without any protections, but it won't be so easy behind a glass window. This is because:

A. Glass is transparent to visible light but opaque to most ultraviolet (UV) radiation.

B. Glass is a poor conductor of heat.

C. Glass acts as a blackbody that absorbs UV and re-emits it as visible light.

D. The glass window reflects all incoming solar radiation like a mirror, protecting your skin.

E. Sunburn is caused by infrared (IR) radiation, which is blocked by the thermal mass of the glass.

Answer: A. Glass is specifically opaque to UVB, the type of UV most responsible for direct sunburn.

(c) **Passive solar heating.** A passive solar house has an interior temperature of 15°C, and heats up at the rate of 120 W per °C temperature difference between the house's interior and the outside. What's the outside temperature if the house absorbs 600 W of solar energy?

A. 0 °C

B. 10 °C

C. 15 °C

D. 20 °C

E. 30 °C

Answer: D. The temperature difference is $\frac{600\ \text{W}}{120\ \text{W/°C}} = 5\ °\text{C}$.

(d) **Solar cost.** The capacity factor (CF) of a PV system is the product of photovoltaic efficiency and the fraction of time that the sunlight can be harvested. A photovoltaic system has an installation cost of \$3.00 per W and a CF of 8%. The cost per kWh averaged over an operational period of 10 years is:

A. \$0.43

B. \$43.0

C. \$0.86

D. \$10.32

Answer: A. The CF tells us the fraction of solar energy we actually harvested, so we have to divide the installation cost by the CF to get the actual cost

$$\frac{\$3.00/\text{W}}{0.08} = \$37.5/\text{W} \tag{2}$$

There are $10 \times 365 \times 24$ hours in 10 years, so the averaged cost over 10 years is

$$\frac{\$37.5/\text{W}}{10 \times 365 \times 24\ \text{h}} \times \frac{1000\ \text{W}}{1\ \text{kW}} = \$0.43/\text{kWh} \tag{3}$$

(e) **Blackbody spectrum.** Which of the following statements are true about the blackbody spectrum? (Select all that apply)

A. As an object gets hotter, the total power it radiates per unit area increases proportionally to the square of its absolute temperature (T^2).

B. As an object's temperature increases, the peak of its radiation spectrum shifts toward shorter wavelengths (higher frequencies).

C. A "perfect" blackbody is a theoretical object that absorbs all incoming radiation and reflects none.

D. The Sun can be modeled as a blackbody with a peak wavelength in the visible light range.

Answer: B, C and D. A is false because the power per unit area (intensity) follows the Stefan-Boltzmann Law ($I = \sigma T^4$).

3. At a hydroelectric power plant, water drops through a head of 22 m before turning a turbine-generator. The flow rate of water through the generator is 180,000 kg/s. If the generation process is 92% efficient, what's the approximate power output of the generator?

   A. 36 MW

   B. 28 kW

   C. 380 MW

   D. 2.2 GW

   Answer: A. $22\ \text{m} \times 9.8\ \text{m/s}^2 \times 180000\ \text{kg/s} \times 0.92 = 3.57 \times 10^7\ \text{W}$

4. Old McDonald is going high tech. He wants to build a solar farm using PV cells that have a net efficiency of 10%. What area of land will he need to produce 1000 MW of electrical energy? Assume that average insolation is 500 W/m^2.

   A. $2 \times 10^7\ \text{m}^2$

   B. $2 \times 10^8\ \text{m}^2$

   C. $2 \times 10^6\ \text{m}^2$

   Answer: A. $\frac{1000\ \text{MW}}{0.1 \times 500\ \text{W/m}^2} \times \frac{10^6\ \text{W}}{1\ \text{MW}} = 2 \times 10^7\ \text{m}^2$.